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ASEN 3802 -- Lab 3 -- Aerodynamics Lab -- Part 1 & 2

Summary:

---------- PART 1 ---------- Task 1: models a 4 digit NACA airfoil both numerically and with an equi-angular representation. Task 2: Uses vortex panel code provided to find the coefficient of lift for different NACA airfoil cases. Also, we will analyze where the coefficient of lift converges based on the number of panels used, N. Task 3: Using thin airfoil theory equations, the effects of airfoil thickness on overall lift will be evaluated. Task 4: Uses previous results to plot and compare sectional lift coefficient versus varying angle of attack for thin airfoil theory, vortex panel method and experimental results. ---------- PART 2 ---------- Task 1: Created Pradtl's lifting line theory function and recreated Anderson 5.20 plot to validate

```
% Author(s): Graeme Appel, McKenna Coakley, Jake Wzientek, Cullen Watz
```

```
% Last Revised: 4/14/2026
```

Housekeeping

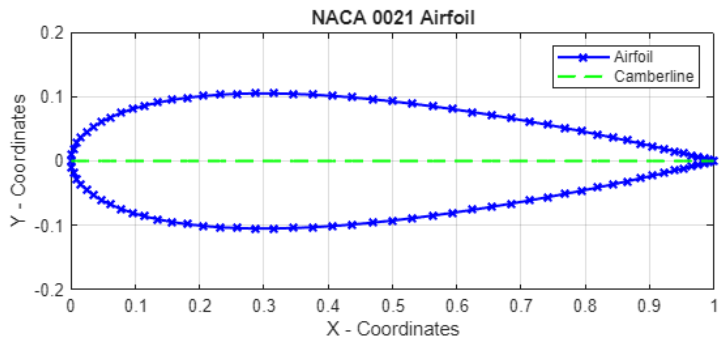
```
clc
clear
close all
```

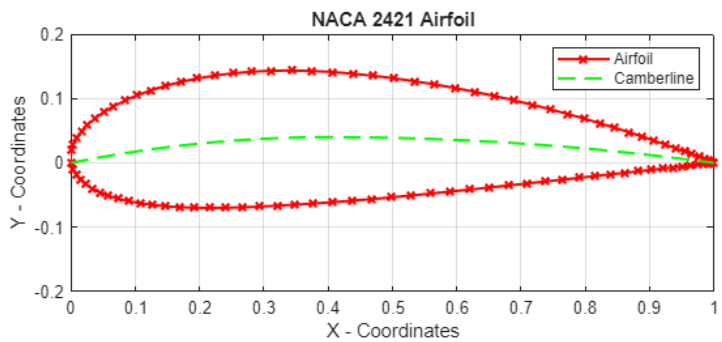
---------- PART 1 ----------

Task 1 -- Plot of NACA 0021 and NACA 2421 using 50 panels

```
% Define Necessary parameters

% NACA 0021
m = 0/100; % max airfoil camber in % of the chord
p = 0/10; % location of max camber measured from leading edge of the chord
measured in 1/10 of the chord
t = 21/100; % max airfoil thickness in % of the chord

% NACA 2421
m2 = 4/100;
p2 = 4/10;
t2 = 21/100;

c = 1; % Chord length (arbitrary units as shape scales regardless of units)
N = 50; % Number of vortex panels employed

% Call Airfoil Distribution function for each case
[x_b, y_b, x_c, y_c] =  NACA_Airfoils(m,p,t,c,N); % reference function 1
below for more details
[x_b2, y_b2, x_c2, y_c2] =  NACA_Airfoils(m2,p2,t2,c,N);

% Plot both airfoil distributions

% NACA 0021 Airfoil
figure();
subplot(2, 1, 1);
plot(x_b, y_b, "-xb", "LineWidth",1.5)
hold on
plot(x_c, y_c, "--g", "LineWidth",1.5)
axis equal % implement correcting dimensions so that MATLAB does not stretch
the y-axis
xlim([0 c]) % set correct graph dimensions
ylim([-0.2*c 0.2*c])
grid on
xlabel('X - Coordinates');
ylabel('Y - Coordinates');
title("NACA 0021 Airfoil")
legend('Airfoil', 'Camberline')

% NACA 2421 Airfoil
subplot(2, 1, 2);
plot(x_b2, y_b2, "-xr", "LineWidth",1.5)
hold on;
plot(x_c2, y_c2, "--g", "LineWidth",1.5)
axis equal % implement correcting dimensions so that MATLAB does not stretch
the y-axis
```

```
xlim([0 c]) % set correct graph dimensions
ylim([-0.2*c 0.2*c])
grid on
xlabel('X - Coordinates');
ylabel('Y - Coordinates');
title("NACA 2421 Airfoil");
legend('Airfoil', 'Camberline')

print('task1', '-dpng', '-r500');

clc
clear
```





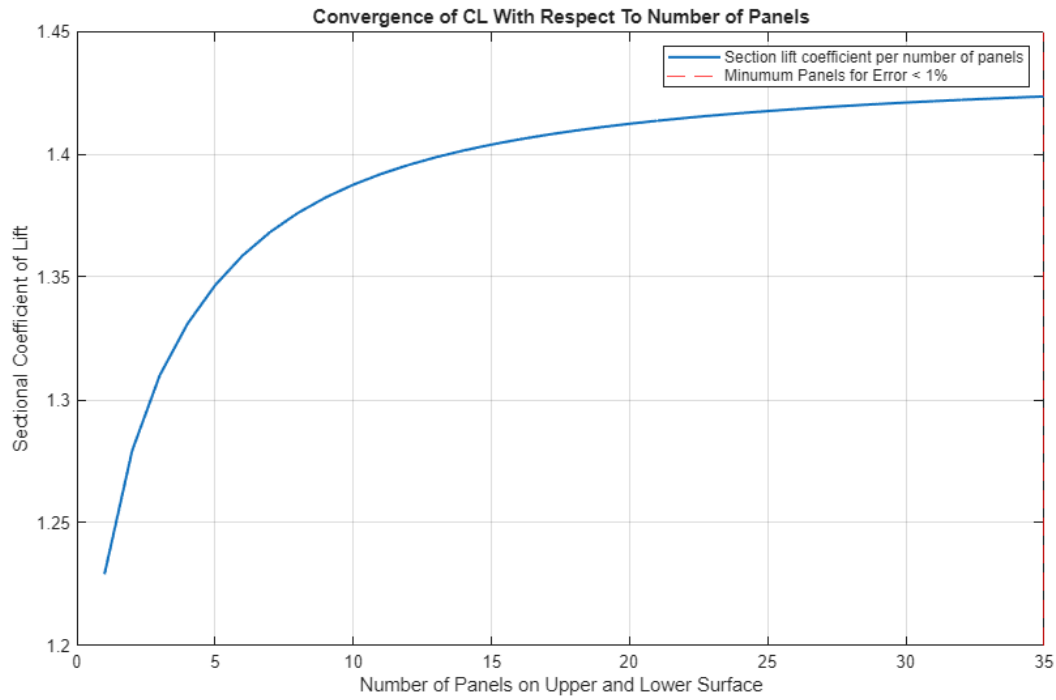
Task 2 -- Applying Vortex Panel Code

Define test variables:

```
i = 1;
m = 0/100;
p = 0/10;
t = 12/100;
c = 1;
CL_old = 0;
error = 1.5;
N_exact = 500;

alpha = 12; % angle of attack in [deg]
v_inf = 50; % freestream velocity in [m/s]
Numerical_Plot = 1; % variable to turn on numerical plots for vortex panel
```

```
method

% Call numerical function to get point distribution
[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N_exact);


% Call vortex panel method code to get output plot
[CL_accurate] = Vortex_Panel(x_b, y_b, v_inf, alpha); % see function 2 below
for more details

N = 2;

% Loop through different vortex panel cases with incrementing N to determine
most ideal amount of
% panels to keep error under 1%.
while error > 1

alpha = 12; % angle of attack in [deg]
v_inf = 50; % freestream velocity in [m/s]
Numerical_Plot = 1; % variable to turn on numerical plots for vortex panel
method

% Call numerical function to get point distribution
[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N);


% Call vortex panel method code to get output plot
[CL(i)] = Vortex_Panel(x_b, y_b, v_inf, alpha);

% error = abs((CL(i)- CL_accurate)/(CL_accurate)) * 100;
error = abs((CL_accurate- CL(i))/(CL(i))) * 100;

% Assign new CL_value for the same error case
if error > 1
    N = N+1;
    CL_old = CL(i);
    i = i + 1;
end
end

N_vec = 1:length(CL); % vector of different vortex panel numbers used --
initialized to same length as CL for use in plotting
disp(['Number of total panels: ', num2str(2*(N))])

% Plot the Number of panels versus the changing/converging CL value
figure();
plot(N_vec, CL,"LineWidth",1.5);
xlabel('Number of Panels on Upper and Lower Surface');
ylabel('Sectional Coefficient of Lift');
title('Convergence of CL With Respect To Number of Panels')
hold on
xline(N_vec(end), "r--") % create line for minimum panels needed for exact
solution
legend("Section lift coefficient per number of panels", "Minumum Panels for
```

```
Error < 1%")
grid on

% Output results to a table
Results_Table = table( ...
    N_exact, CL_accurate, ...
    N, CL(end), error(end), ...
    'VariableNames', {'N_exact', 'CL_exact', 'N_predicted', 'CL_predicted',
'Percent_Error'});
disp(Results_Table)
```



Task 3 -- Applying thin airfoil theory and vortex panel method to analyze accuracy of approximations

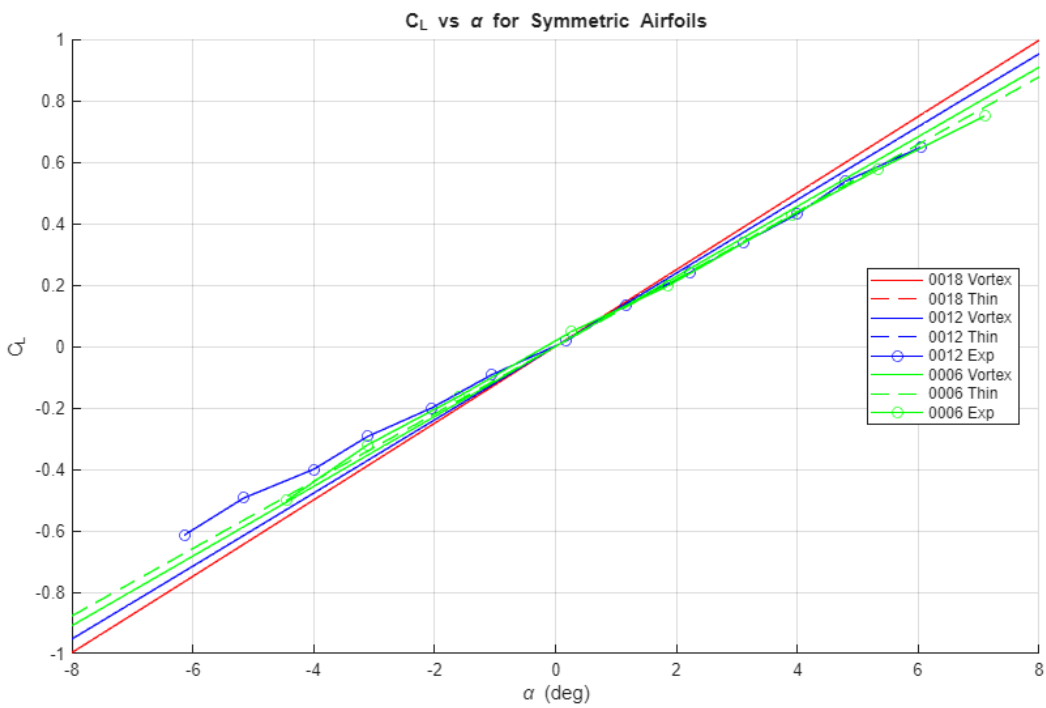
```
% Call thin airfoil theory function for each different airfoil case -->
Note: for each case, alpha
% varies from -8 degrees to 8 degrees

% NACA 0006 case
m = 0/100;
p = 0/10;
t = 6/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector
```

```
% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
% --> See function 3 below for more details
[alpha_L0_thin_0006, alpha_L0_vort_0006, cl_thin_0006, CL_vort_0006,
slope_thin_0006, slope_vort_0006] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);

% NACA 0012 case
m = 0/100;
p = 0/10;
t = 12/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector

% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
[alpha_L0_thin_0012, alpha_L0_vort_0012, cl_thin_0012, CL_vort_0012,
slope_thin_0012, slope_vort_0012] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);


% NACA 0018 case
m = 0/100;
p = 0/10;
t = 18/100;
c = 1;
N_req = N_vec(end); % vortex panels required is the last value of the
converging/increasing N vector

% Call thin airfoil theory function to determine conditions under thin
airfoil theory assumptions
[alpha_L0_thin_0018, alpha_L0_vort_0018, cl_thin_0018, CL_vort_0018,
slope_thin_0018, slope_vort_0018] = Thin_Airfoil_Theory(m, p, t, c, N_req,
alpha, v_inf);


% Read in experimental data experimental data from digitizer for each airfoil

% NACA 0006 data
data1 = load("NACA0006.mat"); % read in data from digitizer
data1 = data1.data;
% Devectorize Data
AoA_exp0006 = data1(:,1);
CL_exp0006 = data1(:,2);

% find L=0 AoA using polyfit
p_0006 = polyfit(AoA_exp0006, CL_exp0006,1);
slope_0006 = p_0006(1); % first coefficient of polyfit yields slope
intercept_0006 = p_0006(2); % 2nd coefficient of polyfit yields y-intercept
AoA_L0_exp0006 = -intercept_0006 / slope_0006;

% NACA 0012 data
```

```
data2 = load("NACA0012.mat");
data2 = data2.data;
% devectorize data
AoA_exp0012 = data2(:,1);
CL_exp0012 = data2(:,2);

% find L=0 AoA using a polyfit
p_0012 = polyfit(AoA_exp0012, CL_exp0012,1);
slope_0012 = p_0012(1);
intercept_0012 = p_0012(2);
AoA_L0_exp0012 = -intercept_0012 / slope_0012;

% NACA 2412 data
data3 = load("NACA2412.mat");
data3 = data3.data;
% devectorize data
AoA_exp2412 = data3(:,1);
CL_exp2412 = data3(:,2);

% find L=0 AoA using a polyfit
p_2412 = polyfit(AoA_exp2412, CL_exp2412,1);
slope_2412 = p_2412(1);
intercept_2412 = p_2412(2);
AoA_L0_exp2412 = -intercept_2412 / slope_2412;

% NACA 4412 data
data4 = load("NACA4412.mat");
data4 = data4.data;
% devectorize data
AoA_exp4412 = data4(:,1);
CL_exp4412 = data4(:,2);

% find L=0 AoA using a polyfit
p_4412 = polyfit(AoA_exp4412, CL_exp4412,1);
slope_4412 = p_4412(1);
intercept_4412 = p_4412(2);
AoA_L0_exp4412 = -intercept_4412 / slope_4412;

% plot CL over alpha for all methods:
figure()
hold on

% Linear angle of attack region is estimated from -8 to 8 degrees
alpha_vec = linspace(-8,8,N_req);

% NACA 0018
plot(alpha_vec, CL_vort_0018, 'r-', 'DisplayName','0018 Vortex')
plot(alpha_vec, cl_thin_0018, 'r--', 'DisplayName','0018 Thin')

% NACA 0012
plot(alpha_vec, CL_vort_0012, 'b-', 'DisplayName','0012 Vortex')
plot(alpha_vec, cl_thin_0012, 'b--', 'DisplayName','0012 Thin')
plot(AoA_exp0012, CL_exp0012, 'b-o', 'DisplayName','0012 Exp')
```

```
% NACA 0006
plot(alpha_vec, CL_vort_0006, 'g-', 'DisplayName','0006 Vortex')
plot(alpha_vec, cl_thin_0006, 'g--', 'DisplayName','0006 Thin')
plot(AoA_exp0006, CL_exp0006, 'g-o', 'DisplayName','0006 Exp')

hold off

xlabel('\alpha (deg)')
ylabel('C_L')
title('C_L vs \alpha for Symmetric Airfoils')
legend('Location','best')
grid on

% Zero Lift Angle of Attack Tables

% NACA 0006 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 0006 (deg)')
Results_Table = table( ...
    alpha_L0_vort_0006, alpha_L0_thin_0006, AoA_L0_exp0006, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0012 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 0012 (deg)')
Results_Table = table( ...
    alpha_L0_vort_0012, alpha_L0_thin_0012, AoA_L0_exp0012, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0018 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 0018 (deg)')
Results_Table = table( ...
    alpha_L0_vort_0018, alpha_L0_thin_0018, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory'});
disp(Results_Table)


% Lift Curve Slope Tables

% NACA 0006 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 0006 (1/deg)')
Results_Table = table( ...
    slope_vort_0006, slope_thin_0006, slope_0006, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 0012 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 0012 (1/deg)')
Results_Table = table( ...
    slope_vort_0012, slope_thin_0012, slope_0012, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
```

```
'Experimental_Data'});
disp(Results_Table)

% NACA 0018 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 0018 (1/deg)')
Results_Table = table( ...
    slope_vort_0018, slope_thin_0018, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory'});
disp(Results_Table)
```



Task 4 (cambered airfoils)

```
%  Call thin airfoil theory function for each different airfoil case


% NACA 2412 (Moderately Cambered)
m = 2/100; p = 4/10; t = 12/100; c = 1;

[alpha_L0_thin_2412, alpha_L0_vort_2412, cl_thin_2412,
CL_vort_2412, slope_thin_2412, slope_vort_2412] =
Thin_Airfoil_Theory(m,p,t,c,N_req,alpha,v_inf);


% NACA 4412 (Highly Cambered)
m = 4/100; p = 4/10; t = 12/100; c = 1;

[alpha_L0_thin_4412, alpha_L0_vort_4412, cl_thin_4412,
CL_vort_4412, slope_thin_4412, slope_vort_4412] =
Thin_Airfoil_Theory(m,p,t,c,N_req,alpha,v_inf);
```

```
% Plot

figure()
hold on

alpha_vec = linspace(-8,8,N_req);

% NACA 0012
plot(alpha_vec, CL_vort_0012, 'b-', 'DisplayName','0012 Vortex')
plot(alpha_vec, cl_thin_0012, 'b--', 'DisplayName','0012 Thin')
plot(AoA_exp0012, CL_exp0012, 'b^', 'LineStyle','none', 'DisplayName','0012
Exp')

% NACA 2412
plot(alpha_vec, CL_vort_2412, 'r-', 'DisplayName','2412 Vortex')
plot(alpha_vec, cl_thin_2412, 'r--', 'DisplayName','2412 Thin')
plot(AoA_exp2412, CL_exp2412, 'r^', 'LineStyle','none', 'DisplayName','2412
Exp')

% NACA 4412
plot(alpha_vec, CL_vort_4412, 'g-', 'DisplayName','4412 Vortex')
plot(alpha_vec, cl_thin_4412, 'g--', 'DisplayName','4412 Thin')
plot(AoA_exp4412, CL_exp4412, 'g^', 'LineStyle','none', 'DisplayName','4412
Exp')

hold off

xlabel('\alpha (deg)')
ylabel('C_L')
title('C_L vs \alpha for Cambered Airfoils')
legend('Location','best')
grid on

% Zero Lift Angle of Attack Tables
% NACA 0012 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 0012 (deg)')
Results_Table = table( ...
    alpha_L0_vort_0012, alpha_L0_thin_0012, AoA_L0_exp0012, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 2412 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 2412 (deg)')
Results_Table = table( ...
    alpha_L0_vort_2412, alpha_L0_thin_2412, AoA_L0_exp2412, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 4412 Airfoil
disp('Table: Zero-Lift Angle of Attack Comparison for NACA 4412 (deg)')
```

```
Results_Table = table( ...
    alpha_L0_vort_4412, alpha_L0_thin_4412, AoA_L0_exp4412, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)


% Lift Curve Slope Tables
% NACA 0012 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 0012 (1/deg)')
Results_Table = table( ...
    slope_vort_0012, slope_thin_0012, slope_0012, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 2412 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 2412 (1/deg)')
Results_Table = table( ...
    slope_vort_2412, slope_thin_2412, slope_2412, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)

% NACA 4412 Airfoil
disp('Table: Lift Curve Slope Comparison for NACA 4412 (1/deg)')
Results_Table = table( ...
    slope_vort_4412, slope_thin_4412, slope_4412, ...
    'VariableNames', {'Vortex_Panel_Method', 'Thin_Airfoil_Theory',
'Experimental_Data'});
disp(Results_Table)
```

```
Table: Zero-Lift Angle of Attack Comparison for NACA 0012 (deg)
    Vortex_Panel_Method    Thin_Airfoil_Theory    Experimental_Data
    ___________________    ___________________    _________________

         -1.86e-15                  0                  -0.18861

Table: Zero-Lift Angle of Attack Comparison for NACA 2412 (deg)
    Vortex_Panel_Method    Thin_Airfoil_Theory    Experimental_Data
    ___________________    ___________________    _________________

          -2.1433                -2.0771               -2.1071

Table: Zero-Lift Angle of Attack Comparison for NACA 4412 (deg)
    Vortex_Panel_Method    Thin_Airfoil_Theory    Experimental_Data
    ___________________    ___________________    _________________

          -4.2779                -4.1543               -3.8067

Table: Lift Curve Slope Comparison for NACA 0012 (1/deg)
    Vortex_Panel_Method    Thin_Airfoil_Theory    Experimental_Data
    ___________________    ___________________    _________________
```

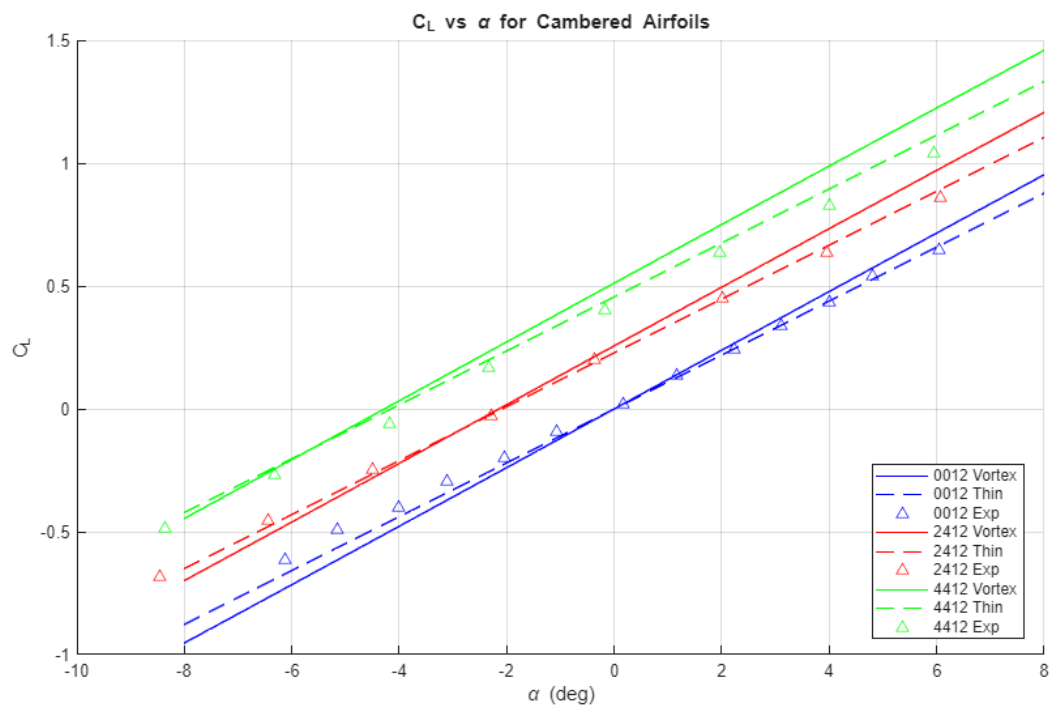
0.11921	0.10966	0.10332

Table: Lift Curve Slope Comparison for NACA 2412 (1/deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
0.11921	0.10966	0.10605

Table: Lift Curve Slope Comparison for NACA 4412 (1/deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
0.11922	0.10966	0.10724



---------- PART 2 ----------

```
% example inputs to test PLLT function
a0_t = 2*pi;
a0_r = 2*pi;
aero_t = 8;
aero_r = 8;
geo_t = 15;
geo_r = 15;
N = 100;
b = 1;
% define specified AR and taper ratio values to replicate Anderson figure
5.20
AR_vec = [4 6 8 10];
```

```
taper_ratio = linspace(0.01,1,100);
% Preallocate delta for plot
delta = zeros(length(AR_vec), length(taper_ratio));

for k = 1:length(AR_vec)
    AR = AR_vec(k);
for i = 1:length(taper_ratio) % compute geometric inputs based on AR and
taper ratio
    c_r = (2*b)./ (AR * (1+taper_ratio(i)));
    c_t = taper_ratio(i) * c_r;
    [e(k,i),c_L,c_Di] =
PLLT(b,a0_t,a0_r,c_t,c_r,aero_t,aero_r,geo_t,geo_r,N);
    delta(k,i) = (1/e(k,i)) - 1;
end
end
figure(); hold on; grid on;
% Anderson 5.20 plot
plot(taper_ratio, delta(1,:), 'LineWidth', 2);
plot(taper_ratio, delta(2,:), 'LineWidth', 2);
plot(taper_ratio, delta(3,:), 'LineWidth', 2);
plot(taper_ratio, delta(4,:), 'LineWidth', 2);
xlabel('Taper Ratio (c_t/c_r)')
ylabel('\delta')
title('Effect of Aspect Ratio and Taper Ratio on \delta')
legend('AR = 4','AR = 6','AR = 8','AR = 10','Location','best')
```

User Defined Functions

1.) NACA Airfoil point distributions and plotting

```
function [x_b,y_b,x_camber,y_camber] =  NACA_Airfoils(m,p,t,c,N)


% Goal/Purpose: Modeling a 4 digit NACA airfoil representation numerically
by converting 4 digit
% inputs into relevant variables and finding discrete x and y coordinates
for the top and bottom of
% the airfoil using numerically derived equations

% Inputs:
% 1.) m = max airfoil camber. This is the 1st digit (leftmost) of the 4
digit NACA airfoil
% representation and represents the max camber of the chord in percentage of
the chord.
% 2.) p = location of max camber. This is the 2nd digit from the left of the
4 digit NACA airfoil
% representation and represents the location of the max camber (measured
from the leading edge) in
% 1/10 of the chord.
% 3.) t = max airfoil thickness. This is the last 2 digits of the 4 digit
NACA airfoil
% representation and represents the maximum thickness of the airfoil in
```

```
percent of the chord.
% 4.) c = total chord length of the airfoil
% 5.) N = # of employed panels for vortex panel method over each surface

% Outputs:
% 1.) x_b = vector containing the x-location of all airfoil boundary points
% 2.) y_b = vector containing the y-location of all airfoil boundary points
% 3.) x_camber = flipped x coordinate distribution of original x coordinate
vector to allow for
% clockwise plotting of airfoil coordinates
% 4.) y_camber = flipped y coordinate distribution of original y_c
coordinate vector to allow for
% clockwise plotting of airfoil coordinates




% Define Thickness Distribution of Airfoil

% Equiangular distribution definition for x-coordinates
for i = 1:N+1
    theta_i = (i-1)*pi/N;             % goes from 0 to pi
    x(i) = c/2 + c/2 * cos(theta_i); % x: c → 0
end

% Define thickness distribution
y_t = ((t * c)/0.2) * ((0.2969 .* sqrt(x./c)) -  (0.1260*(x./c)) -
(0.3516 .* (x./c).^2) + (0.2843 .* (x./c).^3) - (0.1036 .* (x./c).^4)); %
numerical equation given


% Define Camber Line Distribution

% preallocate variables:
y_c = zeros(size(x));
dy_c = zeros(size(x));
zeta = zeros(size(x));

x_u = zeros(size(x));
x_L = zeros(size(x));

y_u = zeros(size(x));
y_L = zeros(size(x));

% Define piecewise function using if statements

for i = 1:length(x)
    if m == 0 || p == 0 % if statement for the case of a symmetric airfoil
with dz/dx = 0

        y_c(i) = 0; dy_c(i) = 0; zeta(i) = 0;

    elseif x(i) < p*c % 1st part of piecewise function
```

```
        y_c(i) = m * (x(i)/(p^2)) * ((2*p) - (x(i)/c)); % camber line
distribution

        dy_c(i) = ((2*m)/p) - ((2*m)/(c*p^2))*x(i); % derivative of camber
line distribution

        zeta(i) = atan(dy_c(i)); % compute zeta

    elseif x(i) >= p*c && x(i) <= c % 2nd part of piecewise function

        y_c(i) = m * ((c-x(i))/(1-p)^2) * (1 + (x(i)/c) - (2*p)); % camber
line distribution

        dy_c(i) = (2*p*m)/(1-p)^2 -((2*m)/(c*(1-p)^2))*x(i); % derivative of
camber line distribution

        zeta(i) = atan(dy_c(i)); % compute zeta

    else

        disp('Wrong x-value');

    end


    % Define X and Y locations over the upper and lower surfaces

        x_u(i) = x(i) - y_t(i)*sin(zeta(i)); % x-coordinates over the upper
surface
        x_L(i) = x(i) + y_t(i)*sin(zeta(i)); % x-coordinates over the lower
surface

        y_u(i) = y_c(i) + y_t(i)*cos(zeta(i)); % y-coordinates over the
upper surface
        y_L(i) = y_c(i) - y_t(i)*cos(zeta(i)); % y-coordinates over the
lower surface

end


    % Group into final output variables

    % Note: output most be clockwise starting and ending at the trailing edge
    % gets rid of duplicate values at leading edge
    x_b = [x_L, flip(x_u(1:end-1))];
    y_b = [y_L, flip(y_u(1:end-1))];
    x_camber = fliplr(x);
    y_camber = fliplr(y_c);


end
```

2.) Vortex Panel Method Code

```
function [CL] = Vortex_Panel(XB,YB,VINF,ALPHA)

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Input:                            %
%                                   %
% XB   = Boundary Points x-location %
% YB   = Boundary Points y-location %
% VINF  = Free-stream Flow Speed    %
% ALPHA = AOA                       %
%                                   %
% Output:                           %
%                                   %
% CL = Sectional Lift Coefficient   %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%
% Convert to Radians %
%%%%%%%%%%%%%%%%%%%%%%%

ALPHA = ALPHA*pi/180;

%%%%%%%%%%%%%%%%%%%%%%
% Compute the Chord %
%%%%%%%%%%%%%%%%%%%%%%

CHORD = max(XB)-min(XB);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Determine the Number of Panels %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

M = max(size(XB,1),size(XB,2))-1;
MP1 = M+1;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Intra-Panel Relationships:                                                  %
%                                                                             %
% Determine the Control Points, Panel Sizes, and Panel Angles                 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    IP1 = I+1;
    X(I) = 0.5*(XB(I)+XB(IP1));
    Y(I) = 0.5*(YB(I)+YB(IP1));
    S(I) = sqrt( (XB(IP1)-XB(I))^2 +( YB(IP1)-YB(I))^2 );
    THETA(I) = atan2( YB(IP1)-YB(I), XB(IP1)-XB(I) );
    SINE(I) = sin( THETA(I) );
    COSINE(I) = cos( THETA(I) );
    RHS(I) = sin( THETA(I)-ALPHA );
end
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Inter-Panel Relationships:              %
%                                         %
% Determine the Integrals between Panels %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    for J = 1:M
        if I == J
            CN1(I,J) = -1.0;
            CN2(I,J) = 1.0;
            CT1(I,J) = 0.5*pi;
            CT2(I,J) = 0.5*pi;
        else
            A = -(X(I)-XB(J))*COSINE(J) - (Y(I)-YB(J))*SINE(J);
            B = (X(I)-XB(J))^2 + (Y(I)-YB(J))^2;
            C = sin( THETA(I)-THETA(J) );
            D = cos( THETA(I)-THETA(J) );
            E = (X(I)-XB(J))*SINE(J) - (Y(I)-YB(J))*COSINE(J);
            F = log( 1.0 + S(J)*(S(J)+2*A)/B );
            G = atan2( E*S(J), B+A*S(J) );
            P = (X(I)-XB(J)) * sin( THETA(I) - 2*THETA(J) ) ...
              + (Y(I)-YB(J)) * cos( THETA(I) - 2*THETA(J) );
            Q = (X(I)-XB(J)) * cos( THETA(I) - 2*THETA(J) ) ...
              - (Y(I)-YB(J)) * sin( THETA(I) - 2*THETA(J) );
            CN2(I,J) = D + 0.5*Q*F/S(J) - (A*C+D*E)*G/S(J);
            CN1(I,J) = 0.5*D*F + C*G - CN2(I,J);
            CT2(I,J) = C + 0.5*P*F/S(J) + (A*D-C*E)*G/S(J);
            CT1(I,J) = 0.5*C*F - D*G - CT2(I,J);
        end
    end
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Inter-Panel Relationships:            %
%                                       %
% Determine the Influence Coefficients %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    AN(I,1) = CN1(I,1);
    AN(I,MP1) = CN2(I,M);
    AT(I,1) = CT1(I,1);
    AT(I,MP1) = CT2(I,M);
    for J = 2:M
        AN(I,J) = CN1(I,J) + CN2(I,J-1);
        AT(I,J) = CT1(I,J) + CT2(I,J-1);
    end
end
AN(MP1,1) = 1.0;
AN(MP1,MP1) = 1.0;
for J = 2:M
    AN(MP1,J) = 0.0;
end
```

```
RHS(MP1) = 0.0;

%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for the gammas %
%%%%%%%%%%%%%%%%%%%%%%%%

GAMA = AN\RHS';

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for Tangential Veloity and Coefficient of Pressure %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for I = 1:M
    V(I) = cos( THETA(I)-ALPHA );
    for J = 1:MP1
        V(I) = V(I) + AT(I,J)*GAMA(J);
    end
    CP(I) = 1.0 - V(I)^2;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Solve for Sectional Coefficient of Lift %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

CIRCULATION = sum(S.*V);
CL = 2*CIRCULATION/CHORD;

end
```

Number of total panels: 72

N_exact	CL_exact	N_predicted	CL_predicted	Percent_Error
500	1.4374	36	1.4235	0.98239

3.) Thin Airfoil Theory Code

```
function [alpha_L0_thin, alpha_L0_vort, cl_thin, CL_vort, slope_thin,
slope_vort] = Thin_Airfoil_Theory(m, p, t, c, N_ideal, alpha,v_inf)

% Goal/Purpose: Find zero lift angle of attack using thin airfoil theory and
computing equation 4.61
% in anderson. Note that we already have dz/dx from previous derivation
input piecewise into our
% NACA_Airfoils function. Then, simply solve for theta from the equation: x
= (c/2)(1-cos(theta))
% and plug this in for theta and numerically integrate.

% Inputs:
% 2.) m = max airfoil camber. This is the 1st digit (leftmost) of the 4
digit NACA airfoil
% representation and represents the max camber of the chord in percentage of
the chord.
```

```
% 3.) p = location of max camber. This is the 2nd digit from the left of the
4 digit NACA airfoil
% representation and represents the location of the max camber (measured
from the leading edge) in
% 3.) t = max airfoil thickness. This is the last 2 digits of the 4 digit
NACA airfoil
% representation and represents the maximum thickness of the airfoil in
percent of the chord.
% 4.) c = chord length of given airfoil configuration
% 1/10 of the chord.
% 4.) N_ideal = ideal number of vortex panels to get within 1% error
% 5.) alpha = arbitrary input angle of attack value(s) in vector form

% Outputs:
% 1.) alpha_L0_thin = zero lift angle of attack computed using thin airfoil
theory
% 2.) alpha_L0_vort = zero lift angle of attack computed using vortex panel
method
% 3.) cl_thin = sectional lift coefficient computed using thin airfoil theory
% 4.) CL_vort = total lift coefficient computed using the vortex panel method
% 5.) slope_thin = lift slope distribition, change in section lift
coefficient versus angle of
% attack computed using thin airfoil theory
% 6.) slope_vort = lift slope distribition, change in section lift
coefficient versus angle of
% attack computed using vortex panel method.

alpha_vec = linspace(-8,8,N_ideal); %create an angle of attack vector
cooresponding to experimental trends in a observable linear CL range
alpha_rad = (pi/180) .* alpha_vec; % convert to radians

% Define Thickness Distribution of Airfoil

% Equiangular distribution definition for x-coordinates
for i = 1:N_ideal+1

    theta_i = (i-1)*pi/N_ideal;             % goes from 0 to pi

    x(i) = c/2 + c/2 * cos(theta_i); % x: c → 0

end


% Define theta_0 in terms of x using chord line equation:

theta_0 =  acos( (1 - x.*(2/c)) );


% Define dz/dx for each part of the piecewise function bounds w.r.t length x


for i = 1:length(x)
    if m == 0 || p == 0 % if statement for the case of a symmetric airfoil
with dz/dx = 0
```

```
    dz_dx(i) = 0;

    elseif x(i) >= 0 && x(i) < p*c  % 1st part of piecewise function

    dz_dx(i) = ((2*m)/p) - ((2*m)/(c*p^2))*x(i); % derivative of camber line
distribution


    elseif x(i) >= p*c && x(i) <= c % 2nd part of piecewise function

    dz_dx(i) = (2*p*m)/(1-p)^2 -((2*m)/(c*(1-p)^2))*x(i); % derivative of
camber line distribution

    else

    disp('Wrong x-value');

    end
end
```

Define zero lift angle of attack with equation 4.61 and numerically integrate

```
% define integrand
integrand = dz_dx.*(cos(theta_0) - 1);

% computing zero lift angle of attack in [rad] w.r.t theta using numerical
integration
alpha_L0_thin = (-1/pi) * trapz(theta_0, integrand);

% loop through for a range of changing alpha values
[x_b, y_b] =  NACA_Airfoils(m,p,t,c,N_ideal);


% loop over all alpha vector values to get vector outputs of CL and cl for
each alpha case
for i = 1:length(alpha_vec)
CL_vort(i) = Vortex_Panel(x_b, y_b, v_inf, alpha_vec(i)); % compute
sectional lift coefficient for vortex panel method case with given alpha
value (note: alpha_L0 should be constant, only alpha input indexes)
cl_thin(i) = (2*pi)*(alpha_rad(i) - -alpha_L0_thin); % compute sectional
lift coefficient for thin airfoil theory case with given alpha value (note:
alpha_L0 should be constant, only alpha input indexes)
end

% calculate lift slope for both vortex and thin airfoil methods:
p_thin = polyfit(alpha_vec, cl_thin, 1); % ensure both alpha inputs and
cl_thin inputs have units of degrees
slope_thin = p_thin(1);

p_vort = polyfit(alpha_vec, CL_vort,1);  % ensure both alpha inputs and
```

```
cl_thin inputs have units of degrees
slope_vort = p_vort(1);
intercept_vort = p_vort(2);
alpha_L0_vort = -intercept_vort / slope_vort; % not sure if this is right.
should AoA L0 be a vector or one value?

alpha_L0_thin = -(180/pi) .* alpha_L0_thin;

end
```

Table: Zero-Lift Angle of Attack Comparison for NACA 0006 (deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
3.4042e-15	0	-0.016067

Table: Zero-Lift Angle of Attack Comparison for NACA 0012 (deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
-1.86e-15	0	-0.18861

Table: Zero-Lift Angle of Attack Comparison for NACA 0018 (deg)

Vortex_Panel_Method	Thin_Airfoil_Theory
1.3556e-14	0

Table: Lift Curve Slope Comparison for NACA 0006 (1/deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
0.11374	0.10966	0.10731

Table: Lift Curve Slope Comparison for NACA 0012 (1/deg)

Vortex_Panel_Method	Thin_Airfoil_Theory	Experimental_Data
0.11921	0.10966	0.10332

Table: Lift Curve Slope Comparison for NACA 0018 (1/deg)

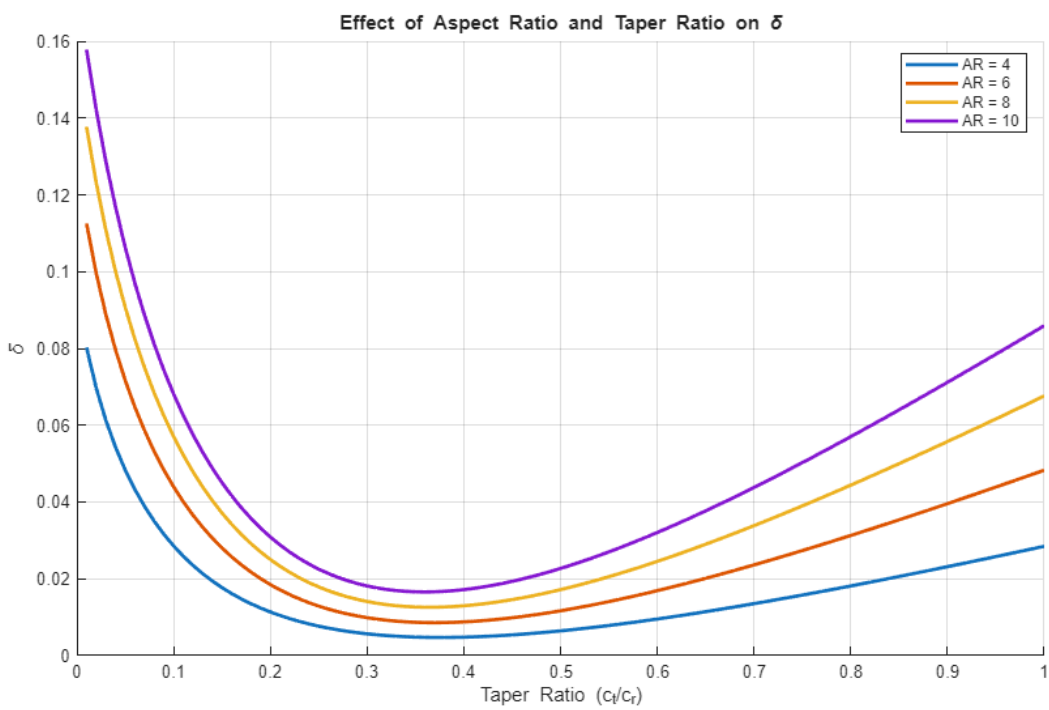
Vortex_Panel_Method	Thin_Airfoil_Theory
0.12474	0.10966

4.) Prandtl Lifting Line Theory Code

Goal/Purpose: Analyze the performance of a finite wing by modeling a sheet of bound horseshoe vortex filaments. It solves the fundamental equation of Prandtl's Lifitng Line Theory for finite wings with thick airfoils to find the fourier coefficients, which can then be used to solve for various performace perameters such as e, CL, and CDi.

```
% Inputs:
% 1.) b = wing span (ft)
% 2.) a0_t = cross-sectional lift slope at the tips (per radian)
% 3.) a0_r = cross-sectional lift slope at the root (per radian)
% 4.) c_t = chord at the tips (in feet)
% 5.) c_r = chord at the root (in feet)
% 6.) aero_t = zero-lift angle of attack at the tips (in degrees)
% 7.) aero_r = zero-lift angle of attack at the root (in degrees)
% 8.) geo_t = geometric angle of attack at the tips (in degrees)
% 9.) geo_r = geometric angle of attack at the root (in degrees)
% 10.) N = number of odd terms to include in the series expansion for
circulation

% Outputs:
% 1.) e = span efficiency factor
% 2.) C_L = coefficient of lift
% 3.) C_Di = induced coefficient of drag

function [e,c_L,c_Di,delta] =
PLLT(b,a0_t,a0_r,c_t,c_r,aero_t,aero_r,geo_t,geo_r,N)
% convert angles of attack to radians
aero_t = aero_t * (pi/180);
aero_r = aero_r * (pi/180);
geo_t = geo_t * (pi/180);
geo_r = geo_r * (pi/180);

% satisfying the fundamental equation at N locations
thetas = (1:N) * pi/(2*N);
% using odd terms
n = 1:2:(2*N-1);
% non-dimensionalize distance along the wing
y = (b/2) .* cos(thetas);
y_normalized = abs(y)/(b/2);
% create linear spans along wing of inputs based on non-dimensionalization
c = c_r + (c_t - c_r) .* y_normalized;
a0 = a0_r + (a0_t - a0_r) .* y_normalized;
AoA_L0 = aero_r + (aero_t - aero_r) .* y_normalized;
AoA_geo = geo_r + (geo_t - geo_r) .* y_normalized;

% [b] = [A][x] where [x] contain fourier coefficients A1, A2 ....
b_vec = (AoA_geo - AoA_L0)';
% i corresponds to theta, j corresponds to N
% generate A matrix using the fundamental equation of PLLT
for i = 1:length(n)
    for j = 1:length(n)
        A_mat(i,j) = ((4*b)/(a0(i)*c(i))* sin(n(j)*thetas(i)) + (n(j) * ...
            (sin(n(j)*thetas(i))/sin(thetas(i)))));
    end
end

% x corresponds to Fourier coefficient column vector
x = A_mat\b_vec;
% solve for delta using solved Fourier Coefficients
delta = 0;
```

```
for i = 2:length(n)
    delta = delta + (n(i) * (x(i)/x(1))^2);
end
% span efficiency factor
e = 1 / (1 + delta);
% planform area
S = (b/2)*(c_r + c_t);
% Aspect ratio
AR = b^2/S;
% Solve for coefficient of lift using first fourier term
c_L = x(1)*pi*AR;

% compute Cdi by summing fourier coefficients
sum = 0;
for i = 1:length(n)
    sum = sum + (n(i) * x(i)^2) * (pi/2);
    c_Di = ((2*b^2)/S) * sum;
end

end
```



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